

Ingenuity is the Name of the Game at Ingenero



Dr. Pratap Nair
President, and CEO
Ingenero

We are thrilled to introduce Dr. Pratap Nair, a distinguished alumnus of IIT Mumbai, where he graduated as a stellar silver medalist. With a doctorate in Chemical Engineering from Rice University, USA, Dr. Nair serves as the Founder Director, President, and CEO of Ingenero. His impressive career spans over three decades, during which he has held leadership roles in renowned global companies such as Cadila and Chemshare Corporation.

Driven by a passion for tackling challenges, Dr. Nair founded Ingenero in 2001 alongside a business partner, transforming it into a leading global entity celebrated for its unmatched expertise and ethical standards in the process manufacturing industry. Ingenero is renowned for its innovative application of AI and cutting-edge engineering practices.

In this candid conversation with Business Today, Dr. Nair shares insights from his remarkable journey and his vision for the future.

What differentiates Ingenero in the industry that it operates in?

What sets Ingenero apart in our industry is our adept application of technology. We leverage AI solutions tailored for precise operations within the Oil and Gas, Midstream, Refining, Petrochemicals, and Chemicals sectors, through our specialized expertise and Frontier AI models. Our unique strength lies in effectively merging technological innovation with engineering expertise, allowing us to navigate the ever-changing and demanding market dynamics with smart strategies. This delicate balance is not just a skill but a challenge that invigorates us, driving our passion to crack complex codes and excel in our field. It brings me immense joy to see how this approach distinguishes Ingenero and

enhances our contribution to the industry.

Could you share about the tech driven expertise by which Ingenero has made a strong presence in the industry?

Our strong presence in the industry is rooted in our ability to add value through our tech-driven expertise. At Ingenero, we specialize in applying AI across the full spectrum of the Oil and Gas, energy, petrochemical, and refining sectors. This specialization enables us to tackle intricate challenges with highly specialized skills and software.

We focus on understanding how to effectively merge diverse synergies like engineering, data science, decision science and IT, to meet and exceed industry demands. Constantly raising our benchmarks is crucial; we prioritize identifying and bridging gaps to drive innovation. I envision the entire Ingenero group as an AI lab, where brainstorming and collaboration are integral to our work culture, fostering the creation of innovative solutions in process manufacturing.

In the realm of Applied AI in Process manufacturing, very few players operate at our level. While others concentrate on developing Foundational AI technology, we are already advancing on the application side, with our Frontier models. By leveraging our domain expertise, we not only address challenges in our domain but also enhance our tools for better applications, ultimately delivering superior results in our field.

In a highly chemical based industry, how do you address sustainability concerns?

Sustainability is a core focus at Ingenero, particularly as it relates to the chemical engineering sector and the pressing need to protect our environment. Our sustainability team plays a critical role in process manufacturing operations, by integrating AI and digital solutions to tackle environmental pollution.

We are committed to significantly reducing emissions through innovative process conversions... In collaboration with esteemed institutions like IIT, we are developing new process designs focused on carbon capture and utilization, hydrogen production, and waste-to-energy technologies. This team is dedicated to addressing sustainability challenges while leveraging our expertise in process engineering.

Our work involves extensive front-end conceptual engineering, where we partner with sustainable technology companies to advance processes from the lab stage to full application. We're also exploring the utilization of renewable energy sources, such as solar energy, to ensure a comprehensive approach to sustainability.

Moreover, we also envision our efforts contributing to the empowerment of rural India. Drawing from my

experience with an Agro division during a previous role, I am eager to engage in initiatives that support sustainable practices in these communities, illustrating our commitment to making a meaningful impact.

What is the global expanse that you are looking at for Ingenero?

We are already a global technology driven products solutions provider. I can confidently say that our technological capabilities at Ingenero are on par with, if not superior to, those in the West. Currently, 70-75 percent of our clientele is based overseas, primarily in the US, the Middle East, and parts of Europe. We serve an international market that recognizes our unique expertise in supporting process manufacturing with advanced technology and analytics—an area where we truly excel.

Unlike many cross-border service providers that operate as low-cost centers in India, we focus on delivering value-added solutions. Our strategy emphasizes the use of advanced technology tools and software systems that provide tangible benefits to our clients. In addition to utilization of hundreds of specialized off the shelf software solutions, many of these tools are also developed in-house, to fill existing gaps, reflecting our commitment to quality and innovation, and they come at a premium because of the value they deliver.

What are the foundational values that sustained you through turbulent times and the way forward?

The foundational values that have sustained me through turbulent times are rooted in my passion for this industry and my commitment to continuous learning. I've been fortunate to engage deeply with the field, gaining exposure to diverse technologies and practices across various geographies. This journey has enriched my understanding of market dynamics, cultural nuances, and the human element, all of which have shaped my perspective.

Learning from mistakes has been invaluable, transforming challenges into opportunities—a lesson instilled in me during my time at IIT. Integrity remains a non-negotiable value for me, even in the face of adversity where others might falter. Upholding this principle fosters a culture that encourages everyone to thrive and positively impacts all stakeholders.

Looking ahead, the future is undeniably focused on technology and AI driven digitalization, areas that resonate naturally with younger generations. As seasoned professionals, it's crucial for us to step out of our comfort zones and engage meaningfully with youth. By empowering them to be self-driven rather than merely directing them, we can unlock their potential and significantly enhance our collective efforts.